SpMM–A2 / GNN aggregation: profiling and controlled design experiment

Weight: 15% of the course grade | Due date: to be announced

Objective

Profile weighted CSR neighbor-feature aggregation on the Xeon Gold 6526Y. Identify how neighbor ordering, feature width and feature-table capacity affect measured performance. The benchmark measures a single-thread kernel, not full GNN inference.

Required investigation

Task	Evidence required
1. Reproducibility — 3 marks	Self-test, source hash, socket/CPU affinity, core/L2/L3 topology, compile flags and checksum.
2. Feature/size sweep — 4 marks	Local vs permuted at F=8 and F=32, R=8,192…262,144 (five repeats each); median ns per edge-feature and layout ratio.
3. Hot function — 2 marks	Use perf record/report to identify csr_spmm; capture source/assembly if available.
4. Memory hierarchy — 3 marks	3 marks: Compare kernel-only base PMU and available L1/L2/LLC/dTLB events. Explain scope, unsupported events and cache-level limitations.
5. Interpretation — 2 marks	Explain how a neighbor's F features affect locality, useful bytes, reuse and cache capacity; give one competing cause and test.
6. Co-design seed — 1 mark	Propose ONE falsifiable software/hardware change (e.g., feature tiling or neighborhood reordering); quantify expected costs before claiming gain.

Scope of working-set sweep

R	X at F=8	X at F=32
8,192	2 MiB	8 MiB
32,768	8 MiB	32 MiB
131,072	32 MiB	128 MiB
262,144	64 MiB	256 MiB

The reference matrix retains 8 nonzeros per row and C=4R. The permuted version uses a deterministic nonlinear bijection. Both layouts must use the same source revision and compute identical logical results.

Exact commands (inside extracted package directory)

```
g++ -O3 -g -fno-omit-frame-pointer -std=c++17 -Wall -Wextra \
  SpMM_GNN_Aggregation_Benchmark.cpp -o spmm_gnn
./spmm_gnn --selftest
python3 Run_SpMM_Experiments.py --reps 5
python3 Plot_SpMM_Results.py results/summary.csv
cat results/summary.csv
```

Hotspot analysis

```
perf list | grep -Ei "L1-dcache|LLC-load|dTLB-load|l2_rqsts"
perf record -g -e cycles:u -o results/profile.data -- \
  ./spmm_gnn --pattern permuted --rows 131072 --nnz-per-row 8 --features 32 --iters 30
perf report -i results/profile.data --stdio > results/hotspots.txt
```

Kernel-only L1 / L2 / LLC / dTLB profiling

```
./spmm_gnn --pmu --pattern permuted --rows 131072 --nnz-per-row 8 --features 32 --iters 30
./spmm_gnn --pmu-levels --pattern permuted --rows 131072 --nnz-per-row 8 --features 32 --iters 30
# Repeat for --pattern local; the runner does both automatically
```

Core and hierarchy counters are collected in separate, freshly initialized processes, both scoped to the timed kernel. L2 uses an Emerald Rapids demand-read true-miss event; no counter equals DRAM bytes. Unsupported or heavily multiplexed events must be reported as N/A. Do not change shared-server perf permissions.

Submission (SpMM_A2_<studentID>)

report.pdf (≤4 pages): normalized performance plots versus feature footprint and width; layout comparisons; L1/L2/LLC/dTLB evidence or N/A; hotspot excerpt; alternative explanation and discriminating test. Attach the complete results/ directory with provenance, raw/summary CSV and logs.

Grading and limitations

This submission carries 15 course percentage points. Assess correctness and reproducibility first; performance explanations must be supported by the collected measurements. Details of the subsequent design stage will be announced separately.